Euler-checked free-group terminals

for compressed normal cuts

A polynomial terminal certificate and an exact homology ledger
Toward quasi-polynomial unknot recognition

Prepared for the ProveIt project
Research and implementation continuation by ChatGPT

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Abstract

The current compressed normal-cut machinery authenticates a finite core and parallelity bundles, but purely geometric column cancellation leaves terminal handlebodies unrecognized. We develop a source-bound algebraic terminal certificate. If every parallelity base is a disk, the whole cut has a finite presentation with at most $74t$ generators, $78t$ relators and $78t$ signed letters, where t is the source tetrahedron count; these active bounds do not depend on the numerical normal weight. A deterministic, nonincreasing singleton Tietze policy has polynomial discovery and independent replay costs.

The topological gate is exact: for an orientable cut with c nonempty-boundary components and certified aggregate free rank r, every component is a handlebody exactly when $\chi = c - r$. No separate irreducibility certificate is needed. If z freely trivial relators were deleted and h_3 source three-handles were retained in the Euler ledger, a completed trace satisfies $b_2 = z - h_3$; already a partial trace with $z > h_3$ proves a whole-cut obstruction. The construction resolves the previously missed figure-eight test cut, with handlebody genera two and one, using genuine signed substitutions beyond initial free reduction and unsigned peeling.

Full proofs, source-bound certificates, an opt-in native implementation, independent verifiers, corpus audits, reproducible measurements, integration instructions and explicit counterexamples to completeness accompany the article. Conditional search-budget theorems identify the remaining global obligations. The generic quasi-polynomial recognition bound is not proved here.

Baseline: `9bacc258bff4b34b4a92b6fa71d290ba5e60ea26`.

Theorems, empirical findings, conditional results and open problems are distinguished throughout. The supplied patch is for review and integration.

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1 Problem, baseline and principal contribution

1.1 What is being accelerated

Unknot recognition asks whether an input knot diagram represents a circle bounding an embedded disk in three-space. The ProveIt programme has developed several exact certificate paths and a substantial compressed normal-surface infrastructure. This continuation concerns the latter: given a validated triangulated source and a supplied normal surface, recognize the homeomorphism types of the components after cutting, without expanding the potentially huge number of parallel normal disks. It addresses a terminal subproblem in a possible hierarchy algorithm. It does not add a new complete scheduler for knot diagrams or certify that an arbitrary supplied triangulation is a particular diagram's exterior.

The distinction is algorithmically significant. A normal vector can have only $7t$ binary coordinates while representing exponentially many disks. An operation polynomial in the number of represented disks can therefore be exponential in input size. The existing native constructors use finite source charts, interval pairings, binary families and source-authenticated orbit certificates to avoid that expansion. The present work uses their output to construct a particularly small *active* group presentation. Large families of isolated balls remain exact integer multiplicities.

The inspected repository is pinned at commit [11]

`9bacc258bff4b34b4a92b6fa71d290ba5e60ea26`

The newest incoming deferred-forests package is present at this revision but its four-module overlay has not been absorbed wholesale into the production API [9]. We preserve that distinction when comparing methods. The new patch applies to this precise production baseline; the deferred comparison runs in an isolated runtime with the four attributed incoming files.

1.2 Why a group terminal is the next useful step

Report 102 established the exact role of unsigned column-leaf cancellation and exhibited geometric stopping sets with a linear deferral requirement [10]. The subsequent incoming work removed the need to enumerate primal forests in the fixed raw positive model and increased whole-cut coverage from 102 to 111 cases on a retained 120-case corpus. It also identified a remaining handlebody cut among the refusals and suggested a free-group terminal under an irreducibility hypothesis [9].

This article makes that suggestion operational and strengthens its logical contract. The maintained source validator does not prove irreducibility. We instead combine exact freeness with the independently authenticated source Euler characteristic. This detects precisely the second homology discarded by algebraic simplification. The resulting theorem avoids both an irreducibility assumption and a separate boundary-component matching step. The finite compiler reads all source two-handles and all disk-bundle annuli; it is a presentation of the *whole cut*, not merely the leftover core.

The new move class uses signs and cyclic words. A defining generator can occur in many other relators, provided it occurs once in its donor; solving for it and substituting is an exact Tietze transformation. Such substitutions can produce cancellations invisible to unsigned incidence. This is useful because the preceding forest theorem already makes repeated primal-tree exploration unproductive inside its stated raw model. Changing the algebraic move class addresses a distinct obstruction.

1.3 Status of the target complexity

Two author-hosted 2021 slide versions state different quantitative forms of the announced quasi-polynomial bound: $n^{O(\log n)}$ in the February slides and $2^{O((\log n)^3)} = n^{O((\log n)^2)}$ in the Oxford handout [4, 5]. Both are quasi-polynomial; their logarithmic exponents should not be silently identified.

The July 2026 paper supplies a hierarchy framework and compressed cutting results. Its Section 9 does not estimate the complete runtime; Proposition 10.2 gives a hierarchy length bounded by $4c_q(\mathcal{H})$, and Theorem 14.4 gives polynomial compressed-cut processing in handle count and logarithmic extended weight [6]. These are valuable ingredients, but do not by themselves supply the required global search bound. We use the announced quasi-polynomial regime as a research target, not as a proved property of the current repository or of this extension.

The older normal-surface complexity results explain why binary compression is the appropriate starting point. Hass–Lagarias–Pippenger provide classical complexity bounds and NP certificates for knot problems [2]. Agol–Hass–Thurston give a polynomial interval-orbit algorithm and compressed normal-surface topology computations [1]. Neither result eliminates the need to control the number of candidate surfaces, the marking carried through a hierarchy, or the number of expensive choices.

1.4 Claim ledger

Table 1 separates the contribution types. The source compiler and checker are ordinary exact Python programs, not a formalized proof in a proof assistant. The mathematical proofs are written below; tests and external comparisons challenge their implementation. No statistical agreement is substituted for a topological theorem.

Result	Status	Scope
Freeness–Euler terminal	Proved from classical topology	Every compact orientable component has nonempty boundary
Disk-base presentation	Proved and implemented	Authenticated source; all bases product disks
Signed singleton reduction	Proved and implemented	Polynomial, one-sided, incomplete
Relator/three-handle ledger	Proved	Exact b_2 on completion; lower bound earlier
Figure-eight improvement	Source-bound, replayed experiment	One supplied cut, not an unknot verdict
Global branch-budget bounds	Conditional theorems	Precise geometric and completeness hypotheses remain open
Generic quasi-polynomial recognizer	Open	Neither the current code nor this article establishes it

Table 1: Logical status of the principal statements.

1.5 Notation and input contract

Let M be the source three-manifold and S the supplied normal surface. Write $W = M \setminus\setminus S$ for the compact manifold obtained by cutting, rather than for the set-theoretic complement without its new boundary. The native input is finite, compact, connected and orientable, with one torus boundary; S satisfies the normal matching and quadrilateral conditions. Thus $\chi(M) = 0$ and

$\chi(W) = \chi(S)$, though the new compiler uses the independently reconstructed additive source Euler summary. Every component of W has nonempty boundary.

We use t for source tetrahedra, h_j for finite core j-handles, p for core ports, a, m, L for active generators, relators and total literal signed word length, c for cut components, and r for the sum of their certified free ranks. All Betti numbers have rational coefficients. A "free-group certificate" proves the entire presented group is free. Computing an abelianization of the expected rank is insufficient.

The output is *unmarked*: a genus census and, when requested, a proof packet. It does not supply meridian disks, peripheral transport, a gluing map to an earlier state, or an essential boundary pattern. In particular a cut of the figure-eight knot exterior can be a union of handlebodies without the figure-eight knot being trivial.

2 A terminal criterion from freeness and Euler characteristic

The preceding terminal proposal assumed irreducibility of every whole-cut component. The source validator does not establish that hypothesis. A different, exact gate removes the need to establish irreducibility and also removes the need for a separate boundary-component census. The topological ingredients below are classical. The contribution to the current programme is their use with a source-authenticated whole-cut presentation and an independently authenticated Euler characteristic.

All manifolds in this section are compact orientable PL three-manifolds. Surfaces and disks are locally flat. Homology without specified coefficients has rational coefficients. Every component of the manifold under consideration is assumed to have nonempty boundary. This last condition holds for the entire normal cut of the current connected compact source manifold with nonempty boundary, even if the normal surface has closed or one-sided components. A complementary component must meet the original boundary or a side of the cutting surface; a component meeting neither would be a closed-and-open component of the original connected manifold.

Lemma 2.1 (Free group and the boundary rank)**.** *Let N be connected, with connected boundary of genus g. If $\pi_1(N) \cong F_g$, then N is a genus-g handlebody.*

Proof. This proof makes its Poincaré-theorem input explicit and does not assume irreducibility. A positive-genus boundary surface cannot have its fundamental group embedded in a free group: subgroups of free groups are free, whereas closed orientable positive-genus surface groups are not. The Loop Theorem therefore supplies a properly embedded compression disk [3, Theorem 3.1]. Cut along such a disk whenever a positive-genus boundary component remains. Van Kampen's theorem expresses the original group as the free product of the cut-component groups, together with an additional free generator when a disk does not separate the manifold. Thus every new component again has free fundamental group.

The disk-cut process terminates because

$$\mu = \sum_{F \subset \partial N,\ g(F) > 0} (2g(F) - 1)$$

strictly decreases under every compression. An essential separating compression decreases this quantity by one; a nonseparating compression decreases it by two, except on a torus, where it decreases it by one. The Loop Theorem is used only as an existence theorem in this argument; the proposed terminal algorithm does not search for these disks.

Suppose that after d disk cuts there are v connected components, with a total of b spherical boundary components and free-group ranks $r_1, \ldots, r_v$. All these components have nonempty

boundary, so $b \geq v$. Every disk compression adds two to the Euler characteristic of the boundary, hence

$$2b = 2 - 2g + 2d, \qquad d = b + g - 1.$$

Reconstructing N attaches d one-handles to the v pieces. The graph of these attachments is connected. Van Kampen and abelianization give

$$g = \sum_{i=1}^{v} r_i + d - v + 1 = g + \sum_{i=1}^{v} r_i + b - v.$$

Every term in $\sum_i r_i + b - v$ is nonnegative. Consequently all r_i are zero and $b = v$. Each terminal component is simply connected and has one sphere boundary. Capping this boundary by a ball gives a closed simply connected three-manifold. The three-dimensional Poincaré theorem [8] identifies it with S^3; the PL Schoenflies theorem, equivalently Alexander's theorem in this dimension, then identifies the uncapped component with a ball [3, Theorem 1.1]. Reversing the disk cuts constructs N from balls and one-handles, so it is a handlebody. Its genus is g. □

Theorem 2.2 (Aggregate free-group–Euler criterion)**.** *Let*

$$W = \coprod_{i=1}^{c} N_i$$

be a finite disjoint union of connected compact orientable PL three-manifolds with nonempty boundary. Form the algebraic free product of the component groups. Suppose

$$*_{i=1}^{c} \pi_1(N_i) \cong F_r.$$

Then the following are equivalent:

1. $\chi(W) = c - r$;
2. $H_2(W; \mathbb{Q}) = 0$;
3. *every N_i is a handlebody.*

When they hold, if $\pi_1(N_i) \cong F_{r_i}$, the genus of N_i is r_i and $\sum_i r_i = r$. In particular, W is a union of balls if and only if, in addition, $r = 0$.

Proof. Each free-product factor injects into F_r. By the Nielsen–Schreier theorem and finite generation, it is a finite-rank free group F_{r_i}. Abelianizing the free product gives $\sum_i r_i = r$. Since every component has nonempty boundary, $H_3(N_i) = 0$, and therefore

$$\chi(W) = c - r + \sum_{i=1}^{c} b_2(N_i). \tag{1}$$

This proves the equivalence of the first two assertions.

Assume $b_2(N_i) = 0$, and write $b_i = |\pi_0(\partial N_i)| \geq 1$. The homology sequence of the pair contains

$$H_3(N_i, \partial N_i) \longrightarrow H_2(\partial N_i) \longrightarrow H_2(N_i) = 0.$$

The first group is one-dimensional, generated by the relative orientation class, and the second has dimension b_i. The first map has rank one: it sends the orientation class to the sum of the boundary orientation classes. Exactness forces $b_i = 1$. Doubling N_i along its boundary gives a closed odd-dimensional manifold, of Euler characteristic zero; thus $\chi(\partial N_i) = 2\chi(N_i)$. If its connected boundary has genus g_i, this identity and $b_2(N_i) = 0$ give

$$1 - g_i = \chi(N_i) = 1 - r_i,$$

so $g_i = r_i$. Lemma 2.1 shows that N_i is a handlebody. The converse follows from the homology of a handlebody. □

There is no need to match two independently produced component histograms in this theorem. It suffices to authenticate the entire free product, the total component count, and the total Euler characteristic. In fact, a group certificate is allowed to mix generators from different algebraic components. If a genus histogram is wanted without a separate boundary census, the implementation should retain the source component of each generator and certify freeness componentwise. That stronger bookkeeping is useful for output, but is not required for the all-handlebody conclusion.

The distinction between group topology and a geometric move is essential. Deleting a freely trivial relator is a Tietze transformation, even though deleting the corresponding two-handle can change the manifold. The Euler gate above accounts for the topology that the fundamental group does not see. The checker must use the source Euler characteristic, not the Euler characteristic of a simplified presentation complex.

Sharpness of the gate. The manifold $S^2 \times I$ has trivial fundamental group, one connected component and Euler characteristic two. It fails $\chi = c - r$, and has two boundary components. Similarly, $H_g \# (S^1 \times S^2)$ has connected genus-g boundary and free fundamental group of rank $g + 1$. Its Euler characteristic is $1 - g$, so its excess over $c - r = -g$ is one. These are genuine non-handlebody obstructions invisible to freeness alone. Conversely, Euler characteristic and abelianization alone cannot replace the free-group certificate; a punctured nontrivial homology sphere is the standard obstruction.

3 A finite whole-cut group compiler for disk parallelity bases

Assume the existing source-parallelity and ordered-interface verifiers have authenticated a normal cut of a t-tetrahedron source, and every parallelity base is a disk. The latter assertion includes orientability, product structure, Euler characteristic one and exactly one boundary component. A ported base has multiplicity one by source authentication. Unported disk bundles may occur with binary-encoded multiplicity. The construction below retains the entire source core, including all its two- and three-handles.

Write h_j for the number of source core j-handles, and κ for the number of components of its zero/one-handle graph. Fix a spanning forest of this graph. Every nonforest one-handle becomes a generator. The cyclic attaching curve of every source two-handle becomes a signed relator by erasing tree crossings. Signs are read from source face flags. Comparing only zero-handle endpoint labels is insufficient when a one-handle is a loop; its two source face flags still distinguish its two directions. Attaching a three-handle along a sphere does not change the fundamental group, so it contributes no relator and is retained in the Euler ledger.

There are two possibilities for a ported disk bundle. Its vertical boundary can be an entire attaching annulus, in which case the ball $D^2 \times I$ is exactly a two-handle attached along $\partial D^2 \times I$. Trace the authenticated annular midpoint through the core and append its signed word as a relator. Alternatively its intersection with the core is a nonempty disjoint union of attaching disks, corresponding to proper interval components of its base boundary. The bundle itself is a ball. It therefore contributes a contractible vertex space, and each attaching disk contributes an edge with trivial edge group in the graph-of-spaces description. Because a disk base has only one boundary component, a full circle and proper interval attachments cannot occur on that same base. An unported disk bundle is an isolated ball.

Let $a_\circ$ be the number of full annular disk bundles, d the number of disk bundles with proper interval attachments, and j the number of those interval attachments. Form a graph Γ with the κ core components and these d disk balls as vertices, and the j attaching disks as edges. If $c_f = |\pi_0(\Gamma)|$, its cycle rank is

$$\beta = j - \kappa - d + c_f.$$

Van Kampen gives the free product of the augmented core groups and an additional free factor of total rank β, with the distribution among components specified by Γ. These additional generators have no occurrences in any relator. This graph calculation uses contractible attaching disks, so no unspecified geometric slide or primitive-curve claim is involved.

Theorem 3.1 (Disk-base whole-cut presentation)**.** *Under the authenticated source hypotheses just stated, the construction produces a presentation of the algebraic free product of the fundamental groups of all whole-cut components. Its numbers of generators and relators are*

$$a = h_1 - h_0 + \kappa + \beta, \qquad m = h_2 + a_\circ. \tag{2}$$

If there are b unported disk balls, counted with their binary multiplicities, the number of whole-cut components is $c = c_f + b$. Furthermore

$$\chi(W) = c - a + m - h_3. \tag{3}$$

For the current native source format there is a uniform linear bound on a, m and total literal signed relator length. The conservative bounds

$$a \le 74t, \qquad m \le 78t, \qquad \sum_{\rho} |\rho| \le 78t$$

suffice. No term depends on the represented normal-surface weight.

Proof. A zero/one-handle core is a regular neighborhood of its graph. Collapsing a spanning forest gives its standard free generators, and the usual handle presentation theorem attaches each two-handle relation using its signed curve. Source flags and the annular-midpoint reconstruction authenticate the precise words. Three-handles do not change the group. The preceding graph-of-spaces argument then proves the presentation claim and (2), including the trivial groups of the isolated balls.

Euler characteristic is additive across these gluings. A full annular disk bundle contributes $\chi(D^2 \times I) - \chi(S^1 \times I) = 1$. A disk bundle with proper interval attachments contributes one, with one subtracted for every attaching disk. Thus

$$\chi(W) = h_0 - h_1 + h_2 - h_3 + a_\circ + d + b - j.$$

Substituting $\beta = j - \kappa - d + c_f$, $c = c_f + b$ and the counts of (2) proves (3).

The existing constructor bounds

$$h_0 \le 6t,\ h_1 \le 8t,\ h_2 \le 12t,\ h_3 \le 4t, \qquad p \le 66t,$$

where p is the number of core ports. The sum of the lengths of all interior edge-link wedge lists is at most $6t$, because each local tetrahedral edge occurs in one such list. Every list is used at most twice, at the two endpoint ranks. The sum of one-handle crossings in all source two-handle attaching cycles is therefore at most $12t$. The ordered-interface verifier partitions the ports into boundary words, each visiting each port once; consequently the additional full annular words have total length at most $p \le 66t$. Erasing forest crossings only decreases length. There are at most p full circle bundles and at most p proper interval attachments. In particular, $a \le h_1 + j \le 74t$ and $m \le h_2 + p \le 78t$. □

The active graph, relators and port incidences have $O(t)$ records, but the compiler must also inspect every supplied base-profile record, including unported disk families. Let f denote the number of those records and let E denote the exact total bit mass of the supplied profile and finite-source data. A conservative structural bound for the delivered forest/ordering implementation is $O(t^2 + f)$ graph and record steps, together with its exact arithmetic cost on

the integers in E. In particular, a minimum-label disjoint-set implementation is not silently assigned an inverse-Ackermann or linear bound. Source preparation and source certificate checking have their separate costs. If the normal coordinates have at most B bits each, the finite chart and source-label description has $O(t(B + \log(t + 2)))$ bits; the finite group presentation itself needs only $O(t \log(t + 2))$ bits after generators are indexed. These statements do not bound the list of profiles or an entire existing orbit certificate by the size of the chart description.

4 Exact polynomial terminal reduction with controlled word length

For this finite group presentation, free and cyclic word reduction may use sign cancellations that the earlier unsigned raw model deliberately excluded. We use only explicit signed words. A current relator ρ admits a defining-generator elimination at x when x or x^{-1} occurs exactly once in ρ, counting both signs. Rotate its cyclic word to

$$\rho = x^{\varepsilon}u, \qquad \varepsilon \in \{1, -1\},$$

where u contains no occurrence of x. The relation is the definition $x = u^{-\varepsilon}$. Substitute this expression for every occurrence of x in every other relator, substitute its inverse for x^{-1}, delete ρ and x, then freely and cyclically reduce the other relators. A freely trivial relator can be discarded; its deletion is recorded in the count z below.

The proposed deterministic producer examines available defining-generator eliminations and accepts one only if the total cyclically reduced relator length does not increase. It stops when every relator is gone or no admitted candidate remains. All source generators, including ones absent from every relator, remain in the generator ledger until explicitly eliminated. A final empty relator list presents the free group whose rank is the number of surviving generators.

Theorem 4.1 (Soundness and polynomial local complexity)**.** *Let a, m be the numbers of source generators and relators, and L the total literal source word length. Use compact integer generator and row labels, or charge the input-label dictionary separately. Put $N = a + m + L + 1$. The procedure above is a sound one-sided freeness test and terminates after at most $\min(a, m)$ defining-generator eliminations. A straightforward exact implementation uses $O(N^4)$ indexed word operations for discovery, $O(N^3)$ for independent replay of a supplied elimination trace, and $O(N^2)$ working letter storage. A trace recording source row labels and eliminated generator labels has $O(m \log(N + 1))$ bits; all substitutions and trivial-relator drops can be reconstructed.*

Proof. Free reduction leaves the represented relator unchanged. Cyclic rotation or cyclic reduction replaces it by a conjugate, so its normal closure is unchanged. A unique occurrence of x makes ρ an explicit definition of that generator. Substitution followed by deleting the definition is an elementary Tietze elimination, with inverse operation adjoining the generator and its defining relation. Deleting a freely trivial relator removes an identity. These facts prove soundness, independently of the length policy. Each elimination removes one generator and one relator, proving termination.

At every accepted state the total reduced relator length is at most L. There are at most L candidate row/generator pairs, since each contains a distinct occurrence within its donor row. For one candidate, the donor tail has length at most L, and at most L letters in all other words are replaced. The total temporary substituted length is at most $O(L^2)$. A stack reduction and endpoint trim reduce those words in linear time in that temporary length. Accounting also for generator and row ledgers costs $O(N^2)$ per candidate. Testing at most N candidates at at most N elimination stages gives $O(N^4)$ discovery work.

A verifier receives the chosen row/generator pair at each stage and needs no candidate search. It reconstructs the donor solution, all substitutions, reductions, empty-relator deletions,

the length gate and the final generator ledger. Thus it uses $O(N^3)$ work. At most $O(N^2)$ temporary letters are required. Stable labels use $O(\log(N+1))$ bits, so the listed pairs have the asserted size. Source authentication and any appended source proof are additional to this small algebraic trace. □

This is a complexity theorem for a specified one-sided producer, not a decision theorem for group freeness. An inconclusive result may reflect the absence of a suitable generator, the length policy, or an unfortunate earlier choice. No order-independence or completeness assertion is made for this greedy signed procedure. The broader certificate format may admit other exactly checkable Tietze moves, but their discovery costs require separate bounds.

Example 4.2 (A free presentation with no initial admitted move)**.** Define an automorphism of $F(a,b)$ by

$$\varphi(a) = ab, \qquad \varphi(b) = a.$$

Its inverse is given by $a \mapsto b$ and $b \mapsto b^{-1}a$, as direct substitution verifies. The two words

$$\varphi^4(a) = abaababa, \qquad \varphi^4(b) = abaab$$

therefore form a free basis. In particular,

$$\langle a, b \mid abaababa, abaab \rangle$$

is the trivial group: applying the inverse automorphism four times takes its relators to a, b. Both displayed words are positive and cyclically reduced. Their (a,b) occurrence counts are $(5,3)$ and $(3,2)$, so neither relator has a generator occurring once. The proposed producer has no initial defining-generator move and returns inconclusive. Adjoining an unused generator c gives the same refusal for a presentation of F_1. This is an exact algebraic counterexample to completeness and needs no group-recognition oracle.

There is nevertheless a precise inclusion at the level of certificates. Every complete unsigned column-leaf trace has a signed Tietze trace that never increases length: its pivot generator occurs once in the entire current relator family, so deleting its donor requires no substitution in another relator. Free and cyclic reductions cannot delete that unique occurrence before its pivot. This proves inclusion of the old certificate class; it does not prove that an arbitrary greedy signed choice preserves the old successful ordering. Retaining the older producer as a fast path makes the inclusion operational.

For a strict algebraic contrast, take

$$\langle x, y \mid xyx^{-1},\ yxy^{-1} \rangle.$$

Its unsigned incidence matrix is

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

whose two rows form a stopping set. Cyclic reduction instead gives the relators y and x, and two defining-generator eliminations certify the trivial group. This is an algebraic illustration; the native experiment provides the separate source-authenticated geometric example.

Corollary 4.3 (Resolving the earlier linear-deferral ball family)**.** *The geometric ball family of report 102 [10], consisting after forest contraction of q blocks with unsigned matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, has a signed terminal certificate with $2q$ defining-generator eliminations, no trivial-relator deletions, and strictly decreasing relator length. Its entire $2q$-row matrix is a raw stopping set, and its old raw deferral number is q. The signed certificate has $O(q \log(q+2))$ bits.*

Proof. Start with one zero-handle and two canceling one/two-handle pairs, a handle presentation of a ball. The initial two-handle words are x, y. Slide the first two-handle over the second along the standard positive band, and then slide the second over the new first. Choose the bands in the zero-handle boundary away from all cocore boundaries. Equivalently, perform these disk slides in the complementary handlebody of the standard genus-two Heegaard splitting of S^3. The complementary disk system remains complete, so the resulting two-handle attachment still gives a ball. This is the geometric realization from report 102; it is not inferred from words alone.

With these positive bands, the two words are xy, yxy. Cyclically rotating the second gives xy^2. They are initially positive and cyclically reduced. Eliminate x using xy, so $x = y^{-1}$; the second relator freely reduces to y. Eliminate y. The total lengths are $5, 1, 0$. There are no empty-relator deletions. Take q copies and form their boundary connected sum along disks disjoint from all attaching regions. The result is a ball. After contracting the connecting zero/one forest its matrix is block diagonal, and its word alphabets are disjoint. The two signed moves therefore apply independently in every block.

Each block's two rows form a stopping set, so the whole $2q$-row set is stopping. Any raw deferral scheme must remove at least one row from each of the q disjoint block obstructions. Conversely, deferring the second row of each block leaves a singleton pivot for the first. Hence the raw deferral number is exactly q, as proved in report 102. The $2q$ signed row/generator choices use $O(\log(q+2))$ bits per move, proving the certificate bound. The new conclusion is the short signed resolution of the earlier geometric obstruction family. □

The family is geometric, but is not asserted to have been exported by the native normal-cut source constructor. The separate native target establishes actual additional coverage in that narrower source class. The displayed block decomposition permits a mathematically linear blockwise construction and replay, but the delivered global solver is only assigned the $O(N^4)$ discovery and $O(N^3)$ replay bounds above. No linear replay implementation is claimed by the block corollary.

Abelianization is insufficient even when an exponent-sum determinant is one. Consider

$$\mathcal{P} = \langle x, y \mid x^3y^{-5},\ (xy)^2x^{-3} \rangle.$$

The exponent-sum matrix is

$$\begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}$$

with determinant one. The group is nevertheless nontrivial: under the usual right-to-left permutation convention, assign

$$x = (1\,2\,3), \qquad y = (1\,3\,5\,2\,4).$$

Then $x^3 = y^5 = 1$ and $xy = (2\,4)(3\,5)$, so both relators vanish in the permutation group, while x has nonidentity image. This exact witness explains why the terminal test certifies the entire group and does not replace it by its exponent-sum matrix.

5 Three-handles give an exact allowance for trivial relators

The source count identity (3) gives a particularly compact interpretation of the Euler gate. It is useful both for implementation and for diagnosing a refusal.

Theorem 5.1 (Relator-deletion ledger and early homology obstruction)**.** *Start with the disk-base whole-cut presentation of Theorem 3.1. Suppose a valid signed trace has performed k defining-generator eliminations and discarded z freely trivial relators, including all such initial normalizations. Then*

$$b_2(W; \mathbb{Q}) \geq z - h_3. \tag{4}$$

If all relators have disappeared, equality holds:

$$b_2(W;\mathbb{Q}) = z - h_3. \tag{5}$$

In particular, a completed free-group trace passes the Euler gate exactly when $z = h_3$. If $z > h_3$ at any intermediate stage, the authenticated whole cut already has positive second Betti number and at least one component is not a handlebody.

Proof. The current presentation has

$$a' = a - k, \qquad m' = m - k - z, \qquad a' - m' = a - m + z.$$

Its abelianization has at most m' independent linear relations on a' generators, so

$$b_1(W;\mathbb{Q}) \geq a' - m' = a - m + z.$$

All components have nonempty boundary; consequently $b_2(W) = \chi(W) - c + b_1(W)$. Insert $\chi(W) = c - a + m - h_3$ to obtain (4).

If $m' = 0$, the remaining presentation is free, of rank $r = a' = a - k$, and $m = k + z$. Substitution into the same Euler formula gives

$$b_2(W) = \chi(W) - c + r = m - k - h_3 = z - h_3.$$

The aggregate criterion of Theorem 2.2 now proves the last conclusions. □

Thus when the core has no three-handles, a complete trace with no freely-trivial relator deletions automatically satisfies the terminal Euler condition. A three-handle is not silently discarded from the manifold: its contribution is recorded by the necessary equality $z = h_3$. A completed trace with $z < h_3$ would contradict the authenticated source hypotheses, because it would give negative b_2; it is not a legitimate alternative terminal topology. A partial trace with $z > h_3$ is a genuine obstruction to the whole cut being a union of handlebodies, but has no direct implication that the original knot is nontrivial: the chosen cut itself may simply be unsuitable for the intended hierarchy.

Binary families cause no growth in this argument. Adding b isolated disk balls adds b to both c and χ, contributes no generator or relator, and leaves every equation above invariant. Only the bit length of b, not its value, enters the arithmetic cost.

Terminal scope. The successful certificate establishes an abstract homeomorphism type and, with componentwise group ledgers, a genus census. It supplies no meridian disk system in marked source coordinates and no transported boundary pattern. Its local polynomial cost does not provide the hierarchy construction, a bounded marked successor, progress, or the global depth bound required for quasi-polynomial unknot recognition.

6 The newly certified native figure-eight cut

The retained fixture `pilot/finite_figureEight/0` provides a concrete source-authenticated example that crosses the preceding cancellation barrier. It contains ten tetrahedra and 36 normal disks. Its finite core has

$$(h_0, h_1, h_2, h_3) = (45, 68, 17, 0).$$

There are five ported parallelity bases, all disks, and every one of their vertical boundaries is attached as a full annulus. The source cut Euler characteristic is -1. The whole-cut compiler produces two presentations, with $(a, m, L) = (7, 5, 18)$ and $(18, 17, 35)$, respectively. Thus the aggregate presentation has 25 generators, 22 relators and 53 letters.

6.1 A complete hand-checkable reduction

For the difficult component, rename the native generators as

$$a = 33,\ b = 41,\ c = 45,\ d = 49,\ e = 57,\ f = 61,\ g = 65.$$

In native row order 17 through 21, the five annular relations are

$$\begin{aligned} &gcf^{-1}b^{-1}f^{-1}d^{-1}, && bed^{-1},\\ &e^{-1}a, && adgc^{-1},\\ &egf^{-1}. \end{aligned}$$

They are already freely and cyclically reduced. Every generator has at least two occurrences across the family, so there is no initial unsigned column leaf. Therefore merely normalizing words before invoking the old peeling routine does not resolve this component.

The following sequence gives a transparent complete algebraic proof:

1. From $e^{-1}a = 1$, eliminate a by $a = e$.
2. From $bed^{-1} = 1$, eliminate b by $b = de^{-1}$.
3. From $egf^{-1} = 1$, eliminate e by $e = fg^{-1}$.
4. The fourth relation becomes $fg^{-1}dgc^{-1} = 1$; eliminate $c = fg^{-1}dg$.
5. The first relation then freely reduces to $gfg^{-1}f^{-1}d^{-1} = 1$; eliminate $d = gfg^{-1}f^{-1}$.

Only f, g survive and no relation remains. Each step solves a generator that occurs once in its donor. The total reduced lengths are

$$18 \longrightarrow 16 \longrightarrow 14 \longrightarrow 11 \longrightarrow 5 \longrightarrow 0.$$

No freely trivial relator is discarded separately. This proves the first component group is F_2 without trusting a group-recognition heuristic.

The deterministic production policy chooses a shorter intermediate trace:

$$(17,41),\ (21,57),\ (18,45),\ (20,49),\ (19,33),$$

where a pair records the donor row and eliminated generator. Its length sequence is

$$18 \longrightarrow 16 \longrightarrow 13 \longrightarrow 7 \longrightarrow 3 \longrightarrow 0.$$

It uses three substitutions involving other relators and then two private column pivots. The independent checker reconstructs every substitution and cyclic reduction from the original source-bound words.

The second component needs seventeen private pivots and leaves one generator, native label 68. Its free rank is one. The aggregate values are consequently

$$c = 2, \qquad r = 2 + 1 = 3, \qquad \chi(W) = -1 = c - r.$$

Theorem 2.2 certifies a genus-two handlebody and a solid torus. Here $h_3 = z = 0$, so the relator ledger agrees automatically.

6.2 What this experiment establishes

The original normal coordinates, source triangulation, complete compressed proof packet and exact row data are included. The public terminal verifier and the dedicated new verifier both replay the new schema. A separately expanded Regina cut has 319 tetrahedra, split into components of 191 and 128 tetrahedra; its recognition gives genera one and two. This comparison supplies an external implementation check, while the delivered certificate verifier uses the finite source charts and theorems above, without invoking Regina.

The incoming deferred audit identifies this fixture as the remaining actual handlebody cut missed by that method [9]. The new result is a strict increase in native whole-cut coverage. It is not an example of recognizing the figure-eight knot as an unknot. It also does not establish that all handlebody cuts have nonincreasing singleton traces: the explicit automorphism counterexample in Section 4 prevents that inference.

7 Certificate architecture and native integration

7.1 Three independently replayed layers

The new implementation separates authenticated geometry, algebraic simplification and the topological terminal gate. The producer writes enough information for a checker to reconstruct each layer. Table 2 lists the six new production modules; two existing entry-point files acquire an opt-in wrapper and schema dispatch.

Module pair in `fastunknot/`	Responsibility
`source_cut_group.py` `source_cut_group_verify.py`	Build/reconstruct the whole-cut disk-base presentation, forests, signs, component partition and source Euler ledger
`free_group_elimination.py` `free_group_elimination_verify.py`	Discover/replay exact nonincreasing singleton Tietze traces
`normal_cut_groups.py` `normal_cut_groups_verify.py`	Bind the native source proofs, presentation, free ranks and Euler-checked genus census

Table 2: The production/checker split. All new production paths use the Python standard library.

The whole certificate has schema `normal-cut-euler-free-v1`. Its `parallelity` field contains the existing authenticated source proof; `boundary` contains ordered native moment-boundary data; `presentation` contains the finite whole-cut presentation; `groups` contains one elimination proof per active cut component; and `census` records the claimed genus histogram and aggregate counts. Isolated disk-ball families retain their exact binary multiplicities.

The source checker reconstructs signed crossings from tetrahedron-face flags, checks continuity of attaching words, and verifies the graph component partition. It checks proposed spanning forests by graph partition and edge count, independently of the producer's forest construction. It checks every core two-handle, every full annular disk attachment and every interval bridge. The source two-handle or annular reader in the producer can be disabled without preventing replay. Authenticated source three-handles contribute to the Euler count even though they have no presentation relator.

The algebra checker uses the original source presentation and the recorded row/generator choices. A SHA-256 digest additionally binds the normalized serialization of the source labels and literal words. Soundness does not rely on hiding source data behind a digest: the whole-cut verifier independently reconstructs and compares the presented source data. The checker verifies the unique occurrence, solves its sign correctly, substitutes in every remaining word,

records identities, enforces the length policy and checks the complete surviving-generator ledger. It does not import the producer.

Finally, the terminal checker authenticates c, r, χ and compares $\chi = c - r$, as well as the equivalent complete-trace condition $z = h_3$. Keeping both is useful as an implementation consistency check. Neither equality licenses a manifold verdict unless all source hypotheses and the complete group trace have already passed.

7.2 API behaviour and failure semantics

The direct entry point is:

```
from fastunknot.normal_cut_groups import normal_cut_group_handlebodies
result = normal_cut_group_handlebodies(
    triangulation, coordinates, record_certificate=True)
```

Its dedicated replay function is `verify_normal_cut_group_handlebodies`. The legacy verifier `verify_normal_cut_handlebodies` dispatches the new schema as well. To retain the geometric path as a fast path, use:

```
from fastunknot.normal_cut_handlebodies import normal_cut_handlebodies
result = normal_cut_handlebodies(
    triangulation, coordinates, group_fallback=True,
    record_certificate=True)
```

The default is `group_fallback=False`; all earlier default producer behaviour and legacy proof schemas remain intact. A geometric success returns immediately. A geometric refusal can invoke the new terminal. This arrangement makes the certificate-class inclusion operational without asserting that the greedy signed policy preserves every possible earlier peeling order.

A source containing a nondisk base, an exhausted orbit budget or a stalled word policy returns INCONCLUSIVE. A completed free trace with positive Euler defect reports `positive_second_betti_number` and its exact aggregate value, while retaining INCONCLUSIVE as its API status. This is an obstruction to this cut being all handlebodies, not a knot decision. The mathematical early $z > h_3$ gate is not yet exported as a separate negative certificate.

The direct producer shares its `max_cycles` allowance across source parallelity queries and its boundary query. The optional two-stage wrapper passes the specified allowance separately to each stage; enabling the fallback therefore permits up to twice that orbit-cycle allowance across the portfolio. This deliberately simple compatibility interface is documented rather than described as a shared global budget. The verifier's `max_operations` counts recorded source orbit operations and Tietze pivots, not all machine instructions. Cooperative callbacks remain the general interruption mechanism. Their exceptions, including `ValueError` and `KeyError`, propagate; they must not be mistaken for malformed-certificate refusals.

7.3 Exact integers and the trusted boundary

The native integer codec supports very large multiplicities through an exact hexadecimal JSON representation. The new path accepts both ordinary in-memory integers and this established serialization. It never coerces a count to floating point. Strict integer checks reject Boolean values in integer fields. An isolated family with multiplicity 2^{16384} is represented by its bit string and contributes directly to c and χ; the compiler never loops over its balls.

There remains shared trusted infrastructure: the Python runtime, basic exact arithmetic, source validation, and the existing independent orbit and ordered boundary verifiers. The new producer/checker pairs reduce common algorithmic failure modes, but do not constitute two

implementations of the entire source geometry stack. Regression tests and explicit Regina comparisons challenge this boundary. The theorem's use of Poincaré is a mathematical implication, not a request for the program to construct Ricci flow or a homeomorphism.

7.4 Why marked geometry must remain separate

A Tietze identity certifies the group. It does not show that a specified attaching curve meets a specified meridian disk once, nor that a disk system can be changed while fixing a peripheral marking. The distinction has concrete precedents in Lackenby's discussion of thickenable presentations and fixed meridian systems [7, Section 6.1]. The present source compiler is geometric, the word simplifier is algebraic, and the Euler theorem recovers only the final unmarked homeomorphism type.

This suggests a disciplined integration point: use the new result to discharge an unmarked terminal query. A hierarchy requiring an essential boundary pattern or a marked successor must separately authenticate that information. Free generators in the output are names in an abstract presentation; interpreting them as source-coordinate meridians would require additional constructions and proofs.

8 Reproducible evidence and performance measurements

8.1 Native corpus and comparison discipline

The final audit uses 106 retained historical source cases and a 14-case extension containing interior vertices and source three-handles. This is a normal-cut corpus, with reducible test manifolds as well as knot exteriors and solid tori. It is not a random sample of knot diagrams or a whole-recognizer benchmark. Every input triangulation and normal vector is copied into the package. Seven cases have binary disk counts beyond the explicit 4096-disk oracle cap; the largest uses a 16385-bit nonzero coordinate.

Three producers are evaluated on each source: the current canonical default, the new direct group terminal, and the independently loaded incoming deferred-forest overlay. All their positive outputs are independently replayed. Table 3 gives the final counts. The two portfolio rows are unions of independently obtained positive certificates; they do not claim a new jointly optimized schedule.

Method	Historical / 106	Interior / 14	Total / 120
Current canonical default	92	12	104
New group terminal alone	64	12	76
Deferred-forest comparison	99	12	111
Canonical plus group	95	12	107
Deferred plus group	100	12	112

Table 3: Fresh source-bound whole-cut handlebody certificates. A refusal is inconclusive.

The older incoming article had 102 canonical positives. At the present pinned revision the default already solves `interior_loop/empty` and `interior_loop/boundary_link`, bringing that baseline to 104. These two are not credited to the new contribution. The new group terminal adds `pilot/finite_trefoil/0`, `pilot/finite_figureEight/0` and `pachner23_5_8_2/mixed` beyond the current default. Only the figure-eight case is new beyond the deferred method.

There are 291 independently replayed positive producer outputs in the audit: $104 + 76 + 111$. All 76 new group positives additionally retain complete serialized packets that were replayed

from their delivered JSON files. A fresh literal Regina [12] cut and component recognition completed on 113 inputs, with no oracle timeout. The remaining seven were deliberately not expanded. The agreement counts are 97 canonical, 69 new group and 104 deferred; the portfolio agrees on all 105 explicit oracle-positive cases. Each of the eight portfolio refusals has an oracle component recognized as nonhandlebody. This describes the finite corpus only and provides no completeness guarantee on unseen sources.

The audit harness initially compared one decoded huge integer with its equal hexadecimal transport string and reported a false disagreement at the 106th case. The integer-field comparison was corrected, and the final 15 cases were run after a hash guard verified all production sources, fixtures and deferred files against the retained first 105. No source or algebraic checker failed. The final report and methodology preserve that recovery rather than presenting the first invocation as uninterrupted. The corrected script can reproduce the whole audit from scratch.

8.2 The Euler gate detects real omitted topology

The disk-base compiler admits 83 of the 120 sources. Of these, 76 obtain a complete handlebody certificate, two stall in the final deterministic word policy, and five have completely free presentations but fail the Euler gate. The other 37 sources contain a nondisk base and are outside this compiler's contract.

For all five Euler failures, the package separately retains and replays the authenticated source presentation and every completed component group trace. Table 4 gives their exact values. The fixture names refer to the saved source cases, and do not assert that a closed manifold itself was fed to the torus-boundary validator. The theorem computes the topology of the corresponding whole cut.

Source fixture	χ	c	r	z	h_3	b_2
`pilot/solid_torus_sum_s2xs1/0`	1	1	1	1	0	1
`interior_2_3/mixed`	3	3	1	2	1	1
`solid_torus_sum_s2xs1/mixed`	3	3	2	2	0	2
`nested_1_4_6/interior_links`	12	7	1	12	6	6
`interior_loop/interior_links`	2	2	1	2	1	1

Table 4: Independently replayed free-group sources with $b_2 = \chi - c + r = z - h_3 > 0$.

These cases justify retaining the original three-handles and source Euler characteristic even when the group simplifier deletes all relators. A terminal rule that identified a free group with a handlebody without the topological gate would give incorrect answers on this retained corpus.

8.3 Independent algebraic audit and regression suite

A separate algebraic audit examines 10203 presentations. It includes all 1365 one-relator presentations on two generators through length five, all 7225 ordered pairs of relators through length three, 1600 deterministic random signed presentations with sparse labels, named nonfree and incomplete free examples, empty edge cases, and the two figure-eight components. The seed and source hashes are retained.

The producer completes 6748 cases and refuses 3455. Each success is checked by an independent explicit-map implementation: it derives the solved generator from the equation $px^{\varepsilon}q = 1$, verifies conjugations used by cyclic reduction, and tracks original generators into the final free group. Original relators must vanish and surviving generators must map identically. Its word oracle repeatedly deletes adjacent inverse pairs rather than importing either production stack reducer.

There are also 481 exhaustive nonabelian S_3 homomorphism-count comparisons, 485 separate Regina group recognitions, and 3240 rejected certificate mutations. For a claimed F_r, the S_3 homomorphism count must be 6^r; this is an additional check, not a proof of freeness. The exact Tietze trace supplies that proof. Both automorphism examples on which the policy stalls are included.

The integrated native regression run passes all 1749 tests. A separate run of the 33 new focused tests passes as well. They exercise source-word orientation, loop one-handles, interval-cycle free factors, three-handles, huge integers, producer-disabled replay, hostile certificates, old default compatibility, budgets and callback propagation. The raw logs are included. No formal proof assistant certification is claimed.

8.4 Measured query cost

Measurements were run after the concurrent regression and audit jobs had finished. Each case uses one warm-up per arm followed by seven randomized-order rounds. Two identical new-terminal arms provide an A/A control. Imports and process startup are outside the measured intervals. Each native arm includes source construction, proof production, JSON serialization/deserialization and independent source replay. Regina includes triangulation and surface import, literal cutting, component recognition and result serialization. Thus the comparison asks for the same unmarked genus census, with the native path also delivering an independently replayed certificate. A refusal is retained as a different outcome and never counted as a faster successful recognition.

The three newly covered native cases test the actual added capability. A one-tetrahedron meridian family separately illustrates dependence on represented weight; that compression benefit already exists in the preceding native infrastructure and is not attributed solely to the new group reducer. Seven rounds provide descriptive local measurements, not a distributional runtime guarantee. Raw samples, A/A ratios, versions and source hashes accompany the derived table.

Case	Group ms	Portfolio ms	Regina ms	Literal tets
Trefoil cut	5.66	5.72	24.71	122
Figure-eight cut	75.46	110.09	22.47	319
Mixed Pachner cut	1251.63	2064.32	3.18	283
One meridian copy	1.83	1.80	0.39	31
64 meridian copies	7.40	7.54	5.87	850
1024 meridian copies	8.40	7.99	111.28	13330
2^{128} copies	8.11	8.63	—	—
2^{500} copies	8.37	8.69	—	—
2^{16384} copies	35.24	36.79	—	—

Table 5: Median complete-query costs in milliseconds, seven rounds. Group uses arm A; the two identical A/A arms are retained in the raw data. Dashes denote the expansion cap. The mixed Pachner row is from the separately retained completed retry.

For the trefoil cut, the new complete certified query takes about 5.7 ms versus 24.7 ms for the literal Regina task. For the figure-eight cut it takes about 75.5 ms versus 22.5 ms. The mixed Pachner case is a substantial regression: about 1.25 seconds for the direct certified query and 2.06 seconds with the failed geometric fast path, versus about 3.18 ms for Regina. Its active group presentation has only 27 letters; the complete source/proof pipeline remains an important practical cost. These comparisons include the different proof deliverables described above and must not be promoted to universal speed claims.

The mixed case exceeded the original aggregate 45-second cap for all warm-ups and

repeated arms. The same seven-round worker was rerun separately and completed; both the original timeout and the completed raw samples are retained. Across the displayed cases, median ratios of the identical A/A arms range from about 0.91 to 1.10, indicating material timing variability. On 1024 meridian copies the direct certified query takes about 8.4 ms versus 111 ms for the literal task, but most of that advantage is inherited binary compression. For 2^{16384} copies the group query is about 35 ms; the explicit task is not attempted, and its cost is not extrapolated.

A single separately retained cProfile run attributes 98.4% of its instrumented wall time to four disjoint inherited stages: parallelity source production, moment-boundary production, and their independent replayers. The new finite compiler, group reducer and corresponding finite checkers together account for about 0.086% under that profiler. Source-proof fields occupy 384724 of the 387809 compact certificate bytes. These are diagnostic attributions, not unprofiled timing estimates. In particular, reducing the 27-letter algebraic trace cannot repair the measured full-query regression by itself.

The profile records 1161 calls both to source interval contraction and to the weighted verifier's prefix-event reconstruction, with 884 translation folds. This suggests two concrete targets: independently checked incremental support and prefix structures, and an explicit orbit path guarded by a polynomial bound on universe size. Appendix C proves that the latter design can preserve binary-input complexity guarantees. Neither source optimization is included in the present production patch.

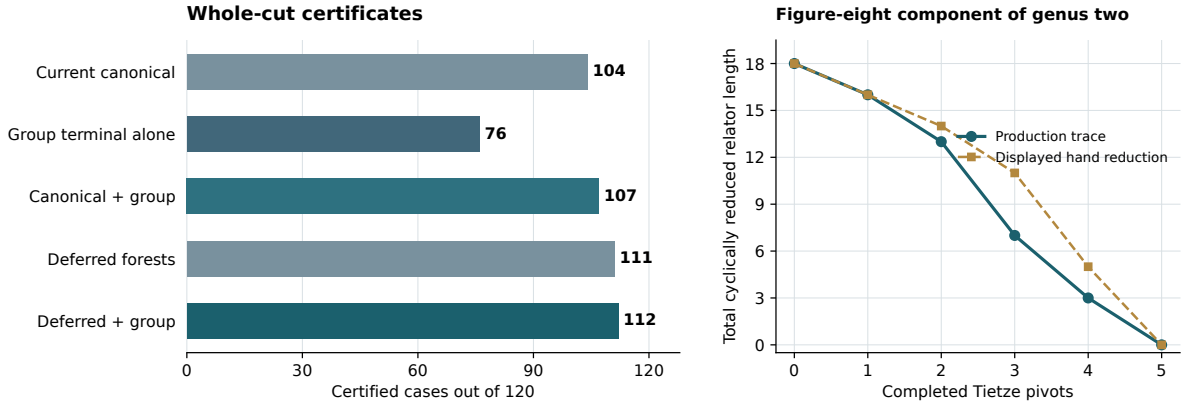


Figure 1: Left: certified whole-cut coverage on the retained 120 cases. Right: exact accepted relator lengths for the difficult figure-eight component. Coverage and word lengths are exact counts, not timing estimates.

The practical integration decision is therefore conservative: preserve the geometric fast path and expose the group terminal as an opt-in fallback. Its main new value is additional certified coverage and a finite algebraic terminal interface. Latency improvements depend on the surrounding scheduler and which expensive alternatives the new certificate avoids.

9 From terminal certification to global complexity

The target considered here is

$$n^{O(\log n)} = \exp\big(O((\log n)^2)\big),$$

where $n \geq 2$ is the original diagram size. The broader term *quasi-polynomial* also permits higher fixed powers of $\log n$ in the exponential. We distinguish these bounds explicitly. None of the

global geometric hypotheses below has been proved for every native hierarchy state in this package.

The new terminal algorithm removes one local obstruction: a source cut with disk parallelity bases can be accepted through a signed group proof even when its unsigned stopping core is large. It does not produce the surfaces leading to that cut, control the number of candidates, or transport a boundary pattern through subsequent cuts. The following results identify precise ways to control those remaining costs.

9.1 Amortized branching, including long repair sequences

Consider the full finite computation tree of a hierarchy search. A node can be a genuine choice of cutting surface, a conjunction requiring several cut pieces to be solved, or a deterministic repair. Count every recursive subproblem, including unsuccessful candidates. Cheap rejected candidates may instead be charged to the node's local work, provided their entire generation and rejection cost is bounded there.

Proposition 9.1 (An amortized branching bound)**.** *Suppose the following conditions hold for inputs of size n.*

1. *The search has a proved sound and complete hierarchy specification. Source creation, candidate generation, admissibility checking, marked transport and terminal discovery and verification cost at most n^C per node and per emitted recursive edge. All required encoded states and local source proofs have at most n^C bits.*
2. *Every root-to-leaf path has at most n^C nodes.*
3. *If $b(v) \geq 1$ is the number of children of a nonleaf node, then every root-to-leaf path P satisfies*

$$\sum_{v \in P \text{ nonleaf}} \log_2 b(v) \leq A(\log_2 n)^2.$$

Then the total search, including unsuccessful candidates, costs $n^{O(\log n)}$ bit operations.

A local sufficient condition for the third hypothesis is a certified integer mass $1 \leq M(v) \leq n^D$, nonincreasing along deterministic edges, for which every branching edge $v \to w$ satisfies

$$\log_2 b(v) \leq K\big((\log_2 M(v))^2 - (\log_2 M(w))^2\big) + e(v,w), \tag{6}$$

where all charges are nonnegative and their sum on each path is $O((\log n)^2)$.

Proof. Assign equal conditional probability $1/b(v)$ to every edge leaving each nonleaf node. A leaf ℓ has probability

$$p_\ell = \prod_{v \in P_\ell} b(v)^{-1} \geq 2^{-A(\log_2 n)^2}.$$

Leaf probabilities sum to one. Hence there are at most $2^{A(\log_2 n)^2}$ leaves. Every node lies on a root-to-leaf path, so the node count is at most the maximum path length times the number of leaves. The edge count is smaller than the node count. Multiplying by the stated local bit cost proves the bound.

For the sufficient condition, sum (6) along a path. The squared-log terms telescope; deterministic edges have zero branching cost and nonnegative potential drop. The initial potential is at most $D^2(\log_2 n)^2$, and the accumulated charges have the stated bound. This gives the required path sum. □

For example, if every branching child has mass at most half its parent and $b(v) \leq M(v)^K$, then (6) holds with zero extra charge. Indeed, with $x = \log_2 M(v) \geq 1$, the potential drop is at

least $x^2-(x-1)^2=2x-1\geq x$. Thus a balanced marked-cut theorem with polynomially many local alternatives is enough, even if polynomially long deterministic repairs occur between its branching steps. The mass need not be the number of represented normal disks; it must be a source-certified quantity with the proved decrease.

The path-length and encoding hypotheses have independent roles. A branching estimate does not bound an arbitrarily long unary computation, and a bounded number of handle families does not bound the bit length of their endpoints or the number of newly split marking intervals. The terminal cost is polynomial when the current t, profile list, integer bit mass and source-proof size are polynomial in n. This fact does not establish the corresponding transport bounds for a successor.

9.2 An exact small terminal key supplied by the new compiler

The signed disk-base compiler has a useful compression property beyond its one-shot runtime. Here W is the authenticated compact orientable PL whole cut, every component has nonempty boundary, and every parallelity base is a disk. Write a, m for its initial generator and relator counts and h_3 for its source three-handle count. The source ledger is

$$\chi(W)=c-a+m-h_3, \qquad r_0:=c-\chi(W)=a-m+h_3.$$

The whole-cut criterion says exactly that W is a union of handlebodies when its component-group free product is free of rank r_0.

Proposition 9.2 (Exact residual terminal kernel)**.** *Apply any independently verified finite Tietze prefix to the compiled presentation. Suppose s surviving generators occur in no remaining relator. Remove these generators from the displayed presentation and call the remaining presentation Q. Put $r_*=r_0-s$. Then*

$$W \text{ is a union of handlebodies} \iff G(Q)\cong F_{r_*}, \tag{7}$$

where the right side is false for $r_*<0$.

Let $k\geq 1$ bound the number of generators, relators and literal letters of Q in total. Impossible target ranks can be rejected immediately. For $0\leq r_*\leq k$, compact relabelling gives a key (Q,r_*) with $O(k\log(k+2))$ bits, and there are $2^{O(k\log(k+2))}$ such keys. Normal-coordinate magnitudes and isolated-ball multiplicities do not occur in this terminal key.

Proof. The full current group is $G(Q)*F_s$. If $G(Q)\cong F_{r_*}$, its free product with F_s has rank r_0, giving the forward terminal conclusion. Conversely, if $G(Q)*F_s\cong F_{r_0}$, then $G(Q)$ is an embedded finitely generated subgroup and hence free. Abelianization shows that its rank is exactly r_0-s. This proves the equivalence.

If the required rank is negative the test is already negative. A free group presented on at most k generators cannot have rank exceeding k, so values outside $0\leq r_*\leq k$ can be separated immediately. Encode the words with row separators over a signed alphabet of at most k generators. The encoded string has $O(k)$ symbols from an alphabet of $O(k)$ symbols, together with the generator count and target rank. This proves both the bit bound and the counting bound. The source compiler's Euler formula shows that isolated balls cancel between c and $\chi(W)$, while a verified Tietze prefix eliminates all dependence on the original geometric labels for this algebraic predicate. □

This is an exact reduction of the terminal predicate, not an algorithm for deciding arbitrary group freeness. The present signed policy is still one-sided. Equal syntactic keys permit reuse of an algebraic certificate after replaying the source compiler and the relabelling map. Each source occurrence still needs its own authenticated binding. A cached greedy refusal remains a refusal of that policy.

The initial active key has $O(t\log(t+2))$ bits. If future normalizations guarantee residual size

$$k\log(k+2) = O((\log n)^2),$$

there are at most $n^{O(\log n)}$ distinct relabelled syntactic terminal keys. In particular $k = O(\log n)$ gives the smaller bound $n^{O(\log\log n)}$. The parameter is the remaining signed presentation size, not the old unsigned deferral number: the earlier q-block geometric stopping family reduces to the empty residual even though its raw deferral is q.

This cache cannot identify hierarchy states. Equal abstract groups and Euler data do not identify peripheral curves, marked faces, or admissible next surfaces. Nor does a small number of terminal keys bound the cost of generating the geometric states that reach them.

9.3 A precise marked-interface route to memoization

There is a distinct route when many search histories give the same future marked problem. The required equivalence is stronger than isomorphism of the terminal groups.

Proposition 9.3 (Source-replayable interface quotient)**.** *Suppose a sound and complete marked hierarchy specification admits keys with the following properties, where* $1 \le k \le n^C$.

1. *A key has at most k active atoms compactly labelled in* $\{1,\dots,k\}$*, at most* $O(k)$ *incidence and cyclic-order records, and at most* $O(k)$ *references to a fixed source dictionary of at most* n^C *exact objects. The dictionary has polynomial total bit mass. Each record has bounded arity and uses only atom labels, dictionary labels and a fixed finite alphabet.*

2. *Equal keys have source-replayable equivalences preserving the entire continuation problem: boundary pattern, orientations, peripheral data, admissibility and the correctness of all cuts. A representative key has at most* n^C *candidate rules with at most* n^C *child keys each, generated and checked in polynomial time. These rules are complete for every source realization of that key.*

3. *Terminal acceptance and its certificate are computed in polynomial time per key, or the terminal certificates are included among the polynomially generated candidate rules. Merely knowing that a short certificate exists does not satisfy this hypothesis. Each terminal proof and each local source equivalence has a polynomial-size, polynomially checkable representation. The source-lifting verifier can compose these objects as a proof DAG in time polynomial in n per DAG node. Source reconstruction, coordinate arithmetic, reindexing and marked transport are included in these bounds; the proof does not require exponentially expanding shared histories.*

Then the hierarchy can be solved in

$$\exp\big(O(k(\log n + \log(k+2)))\big)\,\mathrm{poly}(n)$$

time. In particular $k = O(\log n)$ *yields* $n^{O(\log n)}$.

Proof. There are at most

$$(n^C)^{O(k)}(k+1)^{O(k)} = \exp\big(O(k(\log n + \log(k+2)))\big)$$

keys. Generate the reachable quotient once, reusing every previously seen key. The number of rules and child references is the key count times a polynomial in n.

A rule can require all its children to be accepted, while several rules at one key represent alternative choices. Thus acceptance is the least fixed point of a finite monotone OR/AND rule system. Maintain, for each rule, the number of children not yet accepted; enqueue its parent when that count reaches zero. Each child reference is processed once. Cycles do not certify themselves: a key is accepted only after an accepting terminal or a rule whose children were accepted earlier. Back-pointers therefore give an acyclic accepting proof, even if the explored

quotient contains cycles. The completeness hypothesis identifies this fixed point with the original finite-proof hierarchy problem. The source lifting hypothesis makes the resulting proof DAG valid for the original marked input. All generation, fixed-point and replay costs have the stated bound. □

The common dictionary is a substantive requirement. Its entries may be very large integers, provided there are only polynomially many exact entries and the total bit mass is polynomial. If each of k marked coordinates instead varies freely over B bits, the key bound becomes $2^{O(k(B+\log k))}$. With $k = \Theta(\log n)$ and $B = \Theta((\log n)^2)$, this gives $\exp(O((\log n)^3))$, not the target $n^{O(\log n)}$. Proving that successor endpoints stay in a small dictionary, or proving an equally strong exact normal form, is part of the geometric work.

9.4 The concrete next bridge and the exponent traps

The most direct continuation from the implemented operator is a *marked balanced-cut theorem*: after source-authenticated parallelity compression, either the entire piece has disk bases and a bounded signed terminal certificate, or a polynomially generated, complete family of marked cuts satisfies the mass charge in (6). Every chosen successor must retain polynomial bit size and an exact transport of its inherited pattern. The new finite group compiler enlarges the acceptable terminal class; the missing theorem is the guaranteed arrival in that class or the controlled progress of every recursive alternative. A proof that a particular word policy always succeeds on the guaranteed terminal class would be required; its current Fibonacci counterexample prevents a general freeness-completeness claim.

There are three common accounting failures. Polynomial local work at polynomial branching depth can still cost $2^{n^{\Omega(1)}}$ even with two children per level. A branching factor $n^{O(\log n)}$ at each of $O(\log n)$ levels gives $n^{O((\log n)^2)} = \exp(O((\log n)^3))$. By contrast, multiplying a total call count $n^{O(\log n)}$ by a per-call cost $n^{O(\log n)}$ preserves $n^{O(\log n)}$: exponents add for total work, whereas branch choices multiply along a path. Finally, counting only successfully emitted surfaces omits the potentially dominant cost of unsuccessful surface searches. Both propositions above explicitly charge those searches and require a complete candidate mechanism.

Recheck of the native constants. The bounds $a \le 74t$ and $m, L \le 78t$ remain valid for the present source format. Their word-length argument uses the source validator's exclusion of reverse-edge identifications and its circle-link condition: an interior edge-link traversal visits each local tetrahedral edge occurrence exactly once, so all wedge lists total at most $6t$; at most two endpoint ranks double this to $12t$. The additional annular words use at most the $66t$ authenticated ports once in total. These bounds do not constrain the full base-profile list, integer bit mass, source proof length, transported marking intervals, or the tetrahedron count of later successor states. Those quantities cannot be absorbed into the constants 74 and 78.

10 Further research questions and proposed work programme

The next phase should be driven by precise missing lemmas and adversarial families. A successful experiment is useful when it identifies a parameter or falsifies an attractive universal claim. The figure-eight improvement and the Fibonacci-word refusal together give a concrete starting point: signed substitution matters, but singleton access is not a complete basis-change mechanism. The following questions are ordered roughly from immediate native work to the global geometric bottlenecks.

10.1 Immediate terminal and implementation questions

Research question 10.1 (Recover the two admitted disk-base stalls)**.** Can explicit source-aware Nielsen transformations expose defining generators in the two remaining disk-base word stalls, while keeping a polynomial total word-size budget? First retain their full signed presentations and compare different generator bases. Require exact automorphism certificates and replay of the original words. Success means new whole-cut certificates or a small, repeatable obstruction to the proposed basis policy; a change in abelianized rank alone does not count.

Research question 10.2 (A stronger bounded algebraic policy)**.** What is the smallest additional move family that resolves automorphic images such as $\langle a, b \mid \varphi^k(a), \varphi^k(b)\rangle$ efficiently without making discovery exponential in the rank? Candidate moves include certified Nielsen basis changes, relator multiplication and controlled temporary length growth. Charge both number of moves and intermediate literal or compressed bit size. A short certificate found by unbounded search is a different achievement from a bounded discovery algorithm.

Research question 10.3 (Package the early homology obstruction)**.** Can the partial-trace condition $z > h_3$ be exported as a compact independent whole-cut obstruction certificate? The theorem already proves its soundness. The implementation task is to retain the verified prefix, authenticate the source count and expose a clearly typed result. Such a result can prune a terminal candidate while preserving the possibility that a different normal surface gives a useful hierarchy.

Research question 10.4 (Reuse authenticated source work across the portfolio)**.** Can the geometric and group terminals share one immutable source/boundary packet and one explicit work ledger? The present opt-in wrapper repeats source preparation after a refusal. A useful refactor would reduce measured failed-path latency and eliminate the two-stage orbit allowance, while preserving legacy certificate output and callback semantics. Measure complete successful queries and inconclusive queries separately, including packet serialization.

Research question 10.5 (Incremental relator substitution)**.** Can inverted occurrence lists, lazy substitutions or straight-line programs reduce the current $O(N^4)$ discovery bound and its constants without changing the selected minimum-length move? An occurrence index can avoid scanning words that do not contain the pivot, but exact cyclic cancellation and candidate comparison still need a proof. Record total changed-word work, transient word length and certificate replay cost, rather than only counting chosen pivots.

10.2 Larger source classes and geometric certificates

Research question 10.6 (Extend the compiler beyond disk bases)**.** For a compact base surface with boundary, its group is free but its boundary curves have specified words. Can a graph-of-spaces compiler retain these words, cyclic edge groups for full annuli, and trivial edge groups for intervals with an active size controlled by ports and encoded base complexity? A boundary loop of a positive-genus one-boundary surface is not a free generator. Replacing it by one would be unsound. Very large encoded genus may also force a long commutator word unless a justified compressed representation is used.

Research question 10.7 (Terminal residual kernels in real hierarchies)**.** After geometric peeling and signed simplification, is the terminal residual size logarithmic in the original diagram size on a geometrically defined class? The exact semantic kernel in the preceding section makes this a well-posed parameter question. The old linear-deferral ball family no longer obstructs it, since signed elimination solves every block. The Fibonacci basis family shows that raw singleton residual size is sensitive to coordinates. A useful theorem must therefore specify allowed source-certified basis changes.

Research question 10.8 (Recover source-coordinate meridians)**.** When the Euler/free-group terminal certifies H_g, can a separate procedure construct a complete meridian system with bounded compressed descriptions and verified boundary carriers? The current proof uses the Loop Theorem and Poincaré only for existence. Converting an abstract group basis into disks is an additional geometric problem. A first milestone is the explicit figure-eight genus-two component, with independently checked embedded disks and transports to the original source charts.

Research question 10.9 (Essential boundary patterns at an algebraic terminal)**.** Can a bounded marked pattern be checked for the properties needed by a hierarchy once the ambient piece is certified to be a handlebody? Merely knowing the genus or the abstract group leaves the pattern's embedding unknown. The desired packet should retain cyclic order, side information, slopes where relevant, and witnesses that rule out violating disks. This is a natural point to connect finite boundary graphs with source-coordinate meridian certificates.

Research question 10.10 (Turn apparent basis obstructions into geometric families)**.** Can the algebraically incomplete Fibonacci-word examples be realized and exported by the native normal-cut constructor with controlled triangulation size? Conversely, do the native disk-annulus sources satisfy an extra restriction that rules out some arbitrary finite-presentation obstructions? Producing actual face pairings and normal coordinates, rather than a plausible attaching-word diagram, is the required experiment.

10.3 The global quasi-polynomial bottlenecks

Research question 10.11 (Find an amortized geometric branch potential)**.** Can a compressed hierarchy admit a nonnegative potential whose decrease pays for $\log b$ at a node with b genuine successor alternatives? Polynomially many deterministic repairs are harmless if they do not replenish this potential. Candidate ingredients include boundary pattern complexity, nonproduct core size, a hierarchy surface's essential topology and the residual group kernel. The local inequality in the branch-budget theorem is a concrete target; mere lexicographic termination does not imply it.

Research question 10.12 (Construct a small marked interface quotient)**.** Is there a source-liftable signature on $O(\log n)$ marked interface records that preserves every relevant continuation and terminal condition? The signature must account for mapping-class information that group rank, component incidence and Euler characteristic forget. For every proposed signature, actively search for two equal-key inputs with different outcomes under the same allowed continuation. A collision of this kind is more informative than a high cache-hit rate on easy inputs.

Research question 10.13 (Bound candidate surfaces, not just one chosen surface)**.** Can the algorithm generate a complete family of geometrically useful normal cuts in polynomial or quasi-polynomial total work at each essential branching stage? Efficiently finding one nonseparating surface does not show it is enough for every later repair. The statement should specify which alternatives cover all successful hierarchies and charge the cost of rejected candidates, coordinate generation and source validation. A short existential hierarchy certificate is not a deterministic search bound.

Research question 10.14 (Control marking growth under repeated cuts)**.** Can every successor be represented in size polynomial in the original diagram length, including all coordinates, interval endpoints, boundary patterns and transport proofs? A core with $O(n)$ handles can still carry enormous boundary word data. The bound must cover the complete state and certificate, not only the geometric core or the new active group presentation.

Research question 10.15 (Connect terminal improvements to a diagram-level schedule)**.** Which expensive current recognizer branches can be avoided when a supplied cut receives the new certificate? Implement a bounded experimental schedule that records diagram-to-exterior provenance, candidate-surface cost, source cuts, all retries and final independent verification. Report added successful knot verdicts and total latency, preserving timeout and inconclusive semantics. The current three added cut successes alone do not predict that outcome.

Research question 10.16 (Formalize the smallest trusted mathematical interface)**.** Can the disk-base graph-of-spaces compiler, signed Tietze trace and Euler count ledger be formalized separately from a full normal-surface library? The finite combinatorial layer is a promising first target. The topological theorem could initially be stated with explicit classical assumptions, then connected to a formal 3-manifold development. This would expose exactly which source geometry invariants must be certified rather than inferred from implementation history.

10.4 Suggested milestones

The first milestone is a reviewed integration of the present opt-in terminal, with the three added native cases and five Euler-obstruction packets kept as permanent regressions. Next, extend the residual algebraic policy and measure its effect on actual source presentations; do not optimize arbitrary group examples without confirming that they model the geometric bottleneck. In parallel, construct source-coordinate marked meridian data for the figure-eight target and evaluate the nondisk graph-of-spaces compiler.

The decisive theoretical milestone is then either a logarithmic amortized branch budget or a complete small marked-state quotient, with polynomial state size and proof lifting. The new terminal makes one local operation finite and cheap in the relevant encoding. It also supplies exact refusal information. Those are useful components of a complexity proof, while the global geometric claims above remain the substantive research problem.

A Package contents, reproduction and integration

A.1 Files and dependencies

The archive is intended to be placed in `docs/incoming` or assigned a new research-report number after review. Its root contains a README, integration instructions, a claim ledger, source provenance, license notes and SHA-256 checksums. The article directory contains this complete LaTeX source, section files, bibliography, reproducible vector figure and compiled PDF. The integration directory contains a unified patch and a complete eleven-file overlay under repository-relative paths.

The production extension uses only the Python standard library. The delivered audit scripts additionally use Regina 7.4 for explicit independent comparisons; the saved run used Python 3.12.14. Regenerating the scientific figure requires Matplotlib. Building the article uses pdfLaTeX and BibTeX, with ordinary mathematical and typography packages. Versions and exact source hashes appear in the result files. The package does not vendor the whole ProveIt runtime or third-party software.

Path	Purpose
`article/Euler_Group_Terminals.tex`	Main article and full proofs
`article/Euler_Group_Terminals.pdf`	Rendered article

Path	Purpose
`integration/euler_group_terminals.patch`	Patch against the pinned production source
`integration/overlay/`	Six new production modules, three new test modules and two updated entry-point modules
`fixtures/`	Exact 106 historical and 14 interior-extension sources
`certificates/group/`	All 76 complete new group-terminal packets
`certificates/figure_eight.json`	Convenient standalone target packet, also used by the independent algebraic audit
`certificates/euler_defects.json`	Five source-bound free-group proofs with exact positive b_2
`comparison/deferred/`	Four unchanged attributed incoming modules used only for the deferred comparison
`results/`	Coverage audit, group audit, regression logs, replay reports, raw benchmark samples and methodology
`tools/`	Portable replay, audit, benchmark and figure-generation commands

A.2 Replay before rediscovery

Apply the patch to a separate checkout at the pinned commit as described in `INTEGRATION.md`. From the unpacked package, set a task-specific shell variable to that checkout's fast runtime:

```
fast_root=/path/to/ProveIt/Topology/UnknotRecognition/fast
python tools/verify_packets.py --fast-root "$fast_root"
python tools/audit_euler_defects.py --fast-root "$fast_root" \
  --replay-only certificates/euler_defects.json \\
  --output /tmp/euler_defect_replay.json
```

The first command uses the existing public whole-cut verifier dispatch, so it checks the new schema through the intended integration interface. It requires no Regina installation and performs no proof discovery. Malformed JSON is rejected before replay, including duplicate keys and floating-point values. The source packets use the established exact integer codec for huge values.

Reproduce the independent algebraic audit with:

```
python tools/audit_freeness.py --fast-root "$fast_root" \
  --target certificates/figure_eight.json \
  --output /tmp/independent_freeness.json
```

This command requires Regina and must run without Python's `-O`, since the research audit uses assertions. It produces the 10203-presentation corpus, source hashes, exact checks and counterexamples. Omitting the target omits its two components and leaves 10201 presentations.

The complete fresh native audit is:

```
python tools/audit_native.py --fast-root "$fast_root" \
  --output /tmp/native_audit.json --packets /tmp/group_packets
```

It creates an isolated runtime for the deferred comparison, uses the four preserved modules and replays those positives as well. It launches literal Regina oracle calls with the documented 4096-disk cap and per-case timeout. The command does not modify the production checkout. Its `-resume` option requires matching source identities; the main report explains the single integer-transport recovery in the saved run.

A.3 Tests and measurements

In the integrated fast runtime, run:

```
python -m unittest tests.test_free_group_elimination \
  tests.test_euler_group_terminal tests.test_source_cut_group -v
python -m unittest discover -s tests -v
```

The saved focused and full runs contain 33 and 1749 passing tests, respectively. Existing fixture directories from the repository are needed for the full suite; the eleven-file overlay alone is not a standalone copy of all ProveIt dependencies.

For descriptive complete-query timings, run after other benchmark and audit jobs have finished:

```
python tools/benchmark_terminals.py --fast-root "$fast_root" \
  --output /tmp/terminal_benchmarks.json
```

The script preserves unsuccessful statuses and case timeouts. It never forms a recognition speed ratio using an inconclusive arm. The one-tetrahedron families exercise binary multiplicity; the new target cases exercise changed terminal coverage. The raw A/A arms help identify timing noise. Neither these few samples nor a log-log chart establishes an asymptotic exponent.

A.4 Build and inspect the article

From the package root:

```
python tools/generate_figures.py
bash article/build.sh
```

The build script changes into its own article directory and runs pdfLaTeX, BibTeX and two final pdfLaTeX passes. The source bibliography uses primary literature and pinned repository links. The supplied PDF was compiled, rendered to page images and visually checked; the final build log is retained. There are no external image downloads or generated raster illustrations required to rebuild the article.

A.5 Merging with the incoming deferred work

The patch is verified against the exact current production baseline. The incoming deferred overlay and this patch both edit `normal_cut_handlebodies.py`; applying one full-file overlay on top of the other would discard logic. Integrate their argument handling and fast-path choice deliberately. Preserve the new public verifier dispatch and the old certificate schemas, then rerun the focused checks, the new target and the combined corpus. The 112-case union is evidence for this combined portfolio's potential coverage; the package does not claim that the incoming `forest_strategy` option is already merged into the new wrapper.

B Additional logical boundaries

Closed components. The nonempty-boundary condition in Theorem 2.2 is essential. The closed manifold $S^1 \times S^2$ has $c = r = 1$ and $\chi = 0$, so it satisfies $\chi = c - r$ while not being a handlebody. The current source validator excludes this input. Reusing the theorem in a broader source API requires establishing the boundary condition for every cut component or adding a separate closed component classification.

Source authentication before rank arithmetic. The same numerical presentation, genus and Euler arrays can be attached to unrelated geometry. All numerical equations in the terminal gate are conditional on the verified presentation actually presenting the whole-cut groups. Neither a checksum nor a plausible genus histogram supplies that topological identification. Signs at loop one-handles, full annular disk attachments, interval bridges, core two-handles and binary isolated balls are all part of the binding.

Certificates versus constructive homeomorphisms. The trace certifies a group isomorphism and the theorem determines the unmarked three-manifold type. It does not produce an explicit PL homeomorphism to a standard handlebody. A downstream algorithm requiring meridian disks, boundary embeddings or a re-triangulated marked successor must request and verify those objects separately.

Historical scope. The compression data model, raw stopping-set analysis and deferred-forest results are attributed to the preceding ProveIt work. The topological ingredients are classical. The contributions claimed here are the particular complete source-bound disk compiler and its active bounds, the Euler-checked signed terminal and exact relator ledger, the residual terminal reduction, the new native certificates, and the explicit conditional research programme. No claim of exhaustive historical priority or a completed quasi-polynomial recognition theorem is made.

C A profile-directed bounded-expansion certificate

The mixed Pachner profile identifies substantial compressed-orbit overhead on small represented universes. Expansion can be a sensible small-case method only if the decision to expand is itself tied to encoded input size. This design lemma gives an explicit safe guard and a complete exact witness. It uses standard graph connectivity, and is proposed as a future source-query adapter rather than an implemented normal-cut backend.

Proposition C.1 (Polynomially guarded explicit orbit replay)**.** *Let q interval pairings generate an equivalence relation on $\{0, \dots, U-1\}$. Each pairing is an affine bijection of integer intervals with slope $+1$ or -1, specified by binary endpoints. Let $N \geq 2$ be the total bit length of the complete query, including any binary piecewise-constant additive weights in explicitly listed finite-dimensional integer vectors (or sparse vectors over explicitly named coordinates). Fix a constant $d \geq 1$.*

When $U \leq N^d$, the exact orbit partition admits a certificate consisting of at most $\max\{U-1, 0\}$ source-witnessed forest edges, using $O(U(\log(U+1) + \log(q+1)))$ bits. Discovery and independent replay, including the total weight of every orbit, have polynomial bit complexity in N for certificates within this size bound. If a separate compressed backend costs $T_{\mathrm{comp}}(N)$, this guarded hybrid has bound $T_{\mathrm{comp}}(N) + \mathrm{poly}_d(N)$; in particular it preserves a polynomial or quasi-polynomial complexity class.

Proof. Create one graph vertex per point and, for every source pairing and every point in its domain, the corresponding undirected relation edge. There are at most qU such edges. A spanning forest can be constructed by a standard union-by-size disjoint-set traversal. Record each accepted forest edge by its source pairing index and domain point; the other endpoint is reconstructed from that pairing. Distinct source indices and points require the stated bit length. The forest has $U - c$ edges, where $c \geq 1$ is its component count when $U > 0$; the empty universe has the empty certificate.

For independent replay, check the domain and affine image of every listed edge, reject loops or repeated-cycle edges in the claimed forest, and compute its components by a separate graph traversal. Then enumerate every source pairing edge and check that its endpoints lie in one forest component. Every forest edge belongs to the source relation, so forest connectivity is

contained in the generated equivalence relation. The last check proves the reverse containment. Thus the partition is exact: neither splitting a true orbit nor merging distinct orbits can pass.

Evaluate the source weight description at each point and add the resulting integer vectors over its certified component. A direct scan of the input weight records at each point already has polynomial cost when $U \leq N^d$. With R input weight records and largest coefficient bit length b, each resulting coordinate sum has $O(b + \log(R+1) + \log(U+1))$ bits, and the number of explicitly named weight coordinates is at most N. All graph work, enumeration and exact arithmetic are consequently polynomial in N for fixed d. The guard is tested on binary integers before allocating any point array. For larger U, the explicit algorithm is not entered at all. □

The proposal still needs a native schema and source adapter for each weighted query and each auxiliary fibre-cover query. The source must authenticate the pairings and weight descriptions just as it does for compressed orbit traces. An explicit witness for a small universe is not permission to expand a huge normal surface. The exponent d is fixed independently of numerical weights; the threshold depends on their encoded bit lengths. Benchmarking both modes on complete source-bound queries would determine a practical threshold. This is a concrete next performance experiment supported by the observed profile and by a proved encoded-complexity guard.

References

[1] Ian Agol, Joel Hass, and William P. Thurston. The computational complexity of knot genus and spanning area. *Transactions of the American Mathematical Society*, 358(9):3821–3850, 2006. https://arxiv.org/abs/math/0205057v2.

[2] Joel Hass, Jeffrey C. Lagarias, and Nicholas Pippenger. The computational complexity of knot and link problems. *Journal of the ACM*, 46(2):185–211, 1999. https://arxiv.org/abs/math/9807016.

[3] Allen Hatcher. Notes on basic 3-manifold topology. https://pi.math.cornell.edu/~Hatcher/3M/3Mfds.pdf. Author-linked notes, Theorem 1.1 (Alexander), printed p. 1; Theorem 3.1 (Loop Theorem), printed p. 56. Historical Poincaré wording in these notes is not used.

[4] Marc Lackenby. Unknot recognition in quasi-polynomial time. https://people.maths.ox.ac.uk/lackenby/quasipolynomial-talk.pdf, 2021. February talk slides, 109 pages; theorem on physical page 14; inspected 12 October 2026.

[5] Marc Lackenby. Unknot recognition in quasi-polynomial time. https://people.maths.ox.ac.uk/lackenby/quasipolynomial-talk-oxford-compressed.pdf, 2021. March Oxford handout, 34 pages; theorem on physical page 7; inspected 12 October 2026.

[6] Marc Lackenby. Incompressible surfaces, hierarchies and unknot recognition. https://arxiv.org/html/2607.23350v1, 2026. arXiv:2607.23350v1, 25 July 2026; Sections 9–10 and Theorem 14.4.

[7] Marc Lackenby. The stable Andrews–Curtis conjecture and thickenable presentations of the trivial group. https://people.maths.ox.ac.uk/lackenby/StableAC1june26.pdf, 2026. Author-hosted manuscript, 1 June 2026; Section 6.1 and Claim 6.7.

[8] Grigori Perelman. Finite extinction time for the solutions to the Ricci flow on certain three-manifolds. https://arxiv.org/abs/math/0307245, 2003. arXiv:math/0307245.

[9] ProveIt project. Deferred forests and exceptional incidence. https://github.com/VladimirReshetnikov/ProveIt/tree/9bacc258bff4b34b4a92b6fa71d290ba5e60ea26/docs/incoming, 2026. Incoming archive ProveIt_Deferred_Forests_2026-10-12.zip; four comparison modules retained unchanged with attribution.

[10] ProveIt project. Rectangular cancellation without search. https://github.com/VladimirReshetnikov/ProveIt/tree/9bacc258bff4b34b4a92b6fa71d290ba5e60ea26/Topology/UnknotRecognition/reports/102, 2026. Research report 102, positive cancellation and linear-deferral ball family.

[11] ProveIt project. UnknotRecognition source snapshot. https://github.com/VladimirReshetnikov/ProveIt/tree/9bacc258bff4b34b4a92b6fa71d290ba5e60ea26/Topology/UnknotRecognition, 2026. Commit 9bacc258bff4b34b4a92b6fa71d290ba5e60ea26, inspected 12 October 2026.

[12] The Regina development team. Regina: Software for low-dimensional topology. https://regina-normal.github.io/, 2026. Version 7.4 used as a separate literal-cut and group oracle; version recorded in the supplied result files.